Wildfire behaviour and danger ratings

E Planas, E Pastor - Fire phenomena and the Earth system: an …, 2013 - Wiley Online Library

… the evolution and **dynamics** of a **wildfire**, namely the way fire … **prediction** systems (eg, see Chapter 3 for a historical **review**). … and referenced **wildland fire** behaviour **model** ever developed…

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Towards data-driven operational **wildfire spread** modeling: A report of the NSF-funded WIFIRE workshop

M Gollner, A Trouve, I Altintas, J Block, R de Callafon… - 2015 - drum.lib.umd.edu

… -empirical rate-of-**spread wildfire model**. Simulations with ABL … -**spread wildfire model** operates on a separate surface **model** … , our **review** will restrict itself to experiments focused on fire …

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Generative algorithms for fusion of physics-based **wildfire spread** models with satellite data for initializing **wildfire** forecasts

B Shaddy, D Ray, A Farguell, V Calaza… - … Intelligence for the …, 2024 - journals.ametsoc.org

… Mandel, 2018: Assimilation of fire perimeters and satellite detections by minimization of the residual in a fire **spread model**. Computational Science–ICCS 2018, Y. Shi et al., Eds., …

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Towards predictive **simulation** of **wildfire spread** at regional scale using ensemble-based data assimilation to correct the fire front position

MC Rochoux, C Emery, S Ricci, B Cuenot… - Fire Safety …, 2014 - publications.iafss.org

… The prototype **simulation** capability is then evaluated in a validation test corresponding to a … providing an optimized **forecast** of the **wildfire behavior**. KEYWORDS: **wildfire**, fire modeling, …

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Operational **Forest-Fire spread** forecasting using the WRF-SFIRE **model**

MP Kale, SS Meher, M Chavan, V Kumar, MA Sultan… - Remote Sensing, 2024 - mdpi.com

… -source WRF-SFIRE **model** has been used to carry out surface **forest fire spread** forecasting in the … Global **forecast** system (GFS)-based hourly forecasted weather **model** data obtained …

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L Sun, C Xu, Y He, Y Zhao, Y Xu, X Rui, H Xu - Forests, 2021 - mdpi.com

… automata with a fixed time step to simulate **forest fire spread** may be limited in its ability to … with an existing **forest fire model** to construct an improved **forest fire spread model**, which …

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Attention-based **wildland fire spread** modeling using fire-tracking satellite observations

Y Zou, M Sadeghi, Y Liu, A Puchko, S Le, Y Chen… - Fire, 2023 - mdpi.com

… Physics-based fire models, as outlined in the referenced **review** paper [13], are based on … We used **model** input settings in our CNN fire **spread** models that were similar to the …

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Fifty years of progress in **wildland fire** modelling: from empirical to fully physical CFD models

D Morvan, G Accary, S Meradji… - Comptes …, 2022 - comptes-rendus.academie-sciences …

… The aim of this short **review** is to present the progress made in **wildland fire** modelling … **model** [18–20]. For operational applications, requiring the **simulation** of **wildfire propagation** at a …

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J Ning, H Liu, W Yu, J Deng, L Sun, G Yang, M Wang… - Forests, 2024 - mdpi.com

… **forest fire spread model** … **model** to predict **forest fire spread** by using FIRER in a specific forest ecosystem. The ecological significance of FIRER is that by using it for fire **spread prediction**…

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Development of a **model** system to predict **wildfire** behaviour in pine plantations

MG Cruz, ME Alexander, PAM Fernandes - Australian Forestry, 2008 - Taylor & Francis

… occasional sustained crown fire **propagation** in denser stands. … 7 **wildfire** propagated mostly as a surface fire with periodic … of a **model** system aimed at predicting the rate of **spread** and …

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